Useful Math Facts

1. **Results Regarding e:** For any number a,

$$e^a = \sum_{i=0}^{\infty} \frac{a^i}{i!} \qquad \text{and} \qquad e^a = \lim_{n\to\infty}\left(1+\frac{a}{n}\right)^n$$

2. **Geometric Series:** If $|r| < 1$ and a is an integer,

$$\sum_{k=a}^{\infty} r^k = \frac{r^a}{1-r}$$

3. **The Binomial Theorem:** For $k = 0, 1, 2, \ldots$,

$$(\lambda_1 + \lambda_2)^k = \sum_{i=0}^{k} \binom{k}{i} \lambda_1^i \lambda_2^{k-i}$$

4. **The Gamma Function:**

$$\Gamma(t) = \int_0^\infty x^{t-1} e^{-x}\, dx$$

For any t, it holds that $\Gamma(t+1) = t\Gamma(t)$, and for $n = 1, 2, \ldots$, $\Gamma(n) = (n-1)!$. Also, $\Gamma(0.5) = \sqrt{\pi}$.

5. **Definite Integrals:**

$$\int_0^\infty e^{-ax}\, dx = \frac{1}{a}, \qquad \int_0^\infty x e^{-ax}\, dx = \frac{1}{a^2}, \qquad \int_0^\infty x^m e^{-ax}\, dx = \frac{\Gamma(m+1)}{a^{m+1}}$$

$$\int_0^\infty e^{-ax^2}\, dx = \frac{1}{2}\sqrt{\frac{\pi}{a}}, \qquad \int_0^\infty x e^{-ax^2}\, dx = \frac{1}{2a}, \qquad \int_0^\infty x^m e^{-ax^2}\, dx = \frac{\Gamma((m+1)/2)}{2a^{(m+1)/2}}$$

$$\int_0^1 t^{x-1}(1-t)^{y-1}\, dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \text{The } \textbf{Beta Function}$$